

Unit 5

CHEMICAL ENERGETICS

55 items · 195 marks

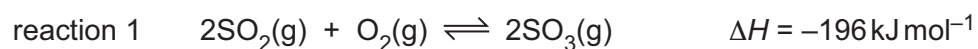
Syllabus sections in this unit

5.1 Enthalpy change, ΔH

5.2 Hess's law

Unit 5: Chemical energetics**1. M/J 26/P23/Q3/a**

- 3** The conditions used for the conversion of sulfur dioxide to sulfur trioxide in the Contact process are carefully controlled.



- (a)** Complete Figure 3.1 to show the reaction pathway diagram for reaction 1.

Draw arrows to represent the enthalpy change, ΔH , and activation energy, E_{A} .

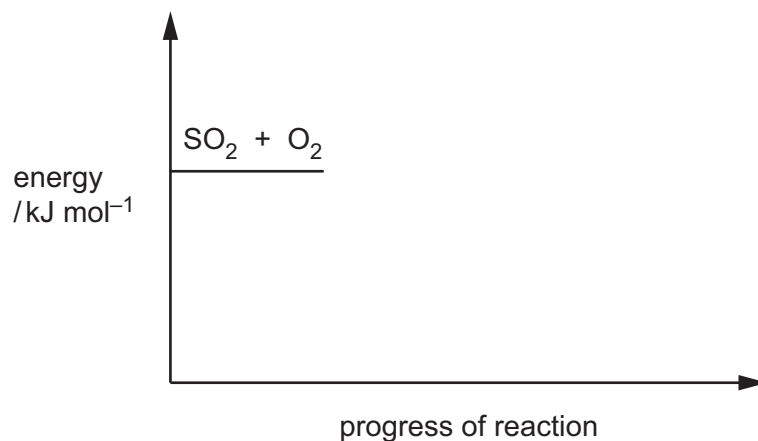


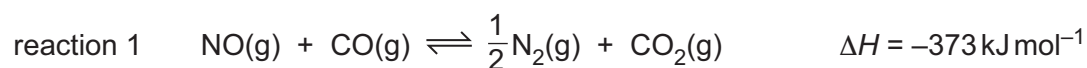
Figure 3.1

[2]

2. MAR 26/P22/Q2/d(i)

2 Nitrogen monoxide, NO, is an atmospheric pollutant.

(d) Reaction 1 occurs during the catalytic removal of NO from car exhaust gases.



The activation energy, E_a , of this reaction is 570 kJ mol^{-1} .

(i) Complete Figure 2.3 to sketch the energy profile for reaction 1.

Show the enthalpy change, ΔH , and the activation energy, E_a .

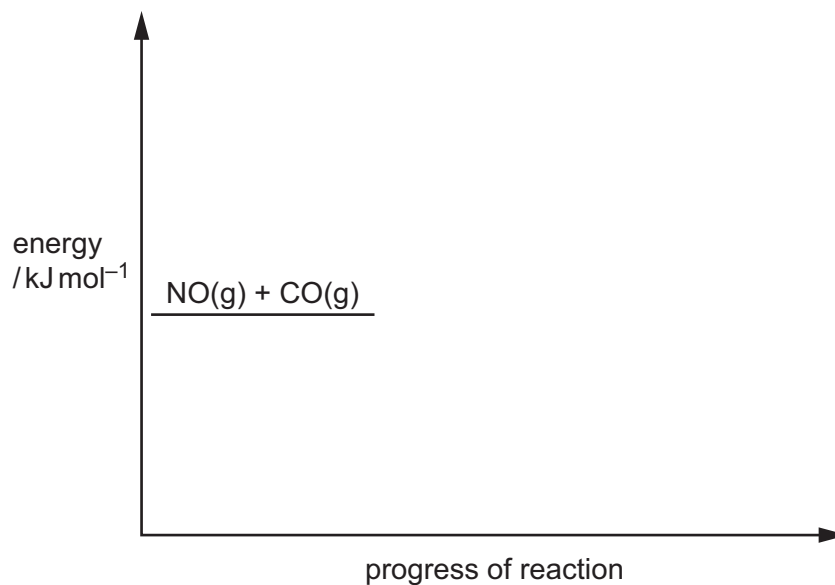


Figure 2.3

[2]

3. O/N 25/P21/Q3/d(ii)

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(d) Fig. 3.2 shows a reaction cycle involving cyclopentane, cyclopentene and $\text{C}_5\text{H}_9\text{Cl}$.

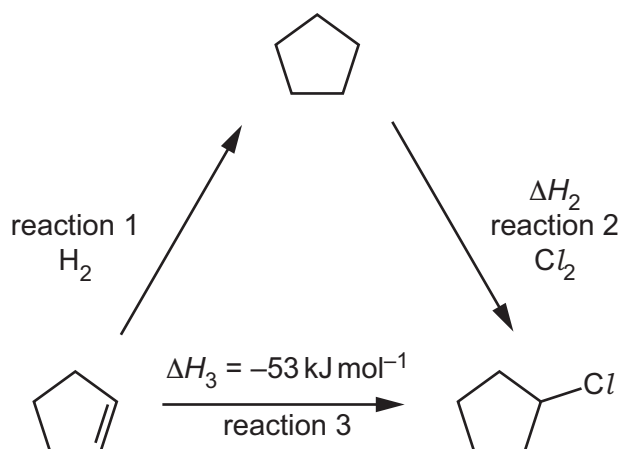


Fig. 3.2

- (ii)** Use the data in Fig. 3.2 and in Table 3.1 to calculate the enthalpy change of reaction 2, ΔH_2 .

Table 3.1

compound	enthalpy change of combustion, $\Delta H_c / \text{kJ mol}^{-1}$
	-3292
	-3115
H_2	-286

$\Delta H_2 =$ kJ mol⁻¹ [2]

4. O/N 25/P22/Q3/e(ii)

3 The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

(e) Hexane can be converted into compounds **E** and **F** at high temperature and pressure. Fig. 3.2 shows the reaction scheme involving hexane, **E** and **F**.

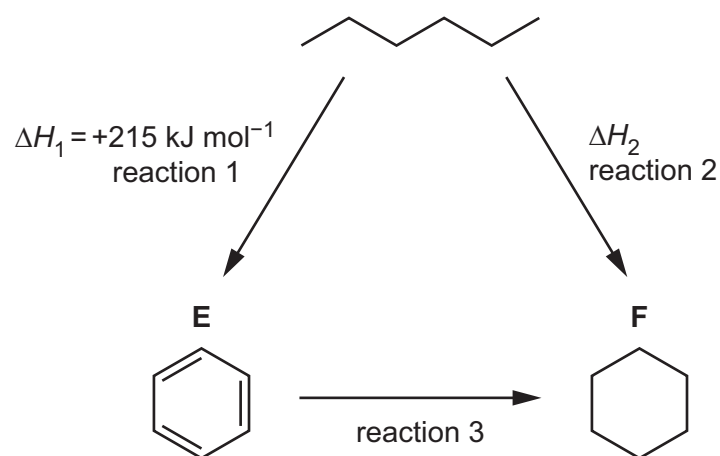
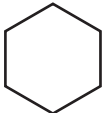
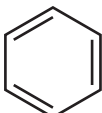


Fig. 3.2

(ii) Use the data in Fig. 3.2 and in Table 3.1 to calculate the enthalpy change of reaction 2, ΔH_2 .

Table 3.1

compound	enthalpy change of formation, $\Delta H_f / \text{kJ mol}^{-1}$
	-156
	+48

$\Delta H_2 =$ kJ mol⁻¹ [2]

5. O/N 25/P23/Q3/d(ii)

- 3 Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(d) Fig. 3.2 shows a reaction cycle involving cyclopentane, cyclopentene and $\text{C}_5\text{H}_9\text{Cl}$.

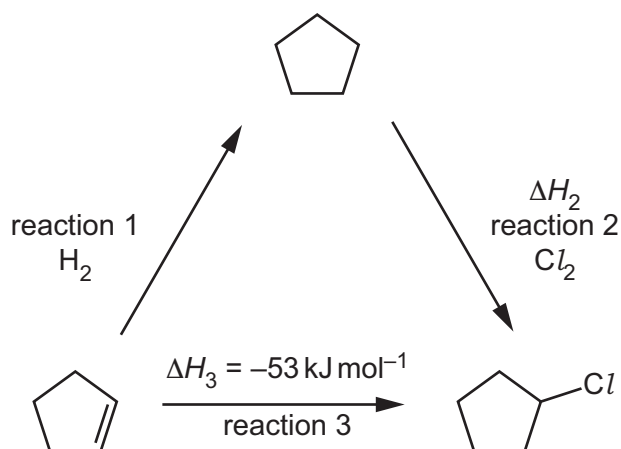


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$\Delta H_2 =$ kJ mol^{-1} [2]

6. O/N 25/P24/Q1/b(iii,iv)

1 The Group 17 elements are oxidising agents.

(b) Cl_2 and Br_2 can each react with NH_3 to give N_2 and a hydrogen halide, HX .



The relative bond strengths of X-X and H-X determine the difference in enthalpy change of the two reactions.

(iii) Define enthalpy change of formation, ΔH_f .

[2]

(iv) Table 1.1 gives data relevant to the reaction of $\text{Cl}_2(\text{g})$ with $\text{NH}_3(\text{g})$.

Table 1.1

compound	enthalpy change of formation, $\Delta H_f^\circ / \text{kJ mol}^{-1}$
$\text{NH}_3(\text{g})$	-46
$\text{HCl}(\text{g})$	-92

Use the data in Table 1.1 to calculate the enthalpy change of the reaction of $\text{Cl}_2(\text{g})$ with $\text{NH}_3(\text{g})$.

enthalpy change of reaction = kJ mol^{-1} [2]

7. M/J 25/P22/Q3/c

(c) Hydrocarbon **Y** contains two $\text{C}=\text{C}$ groups. There are no other functional groups present.

Y reacts with an excess of $\text{H}_2(\text{g})$ to produce hexane, C_6H_{14} .

Table 3.1

bond	energy / kJ mol^{-1}
C-C	350
$\text{C}=\text{C}$	610
$\text{C}\equiv\text{C}$	840
H-H	436
C-H	410

Use Table 3.1 to calculate the enthalpy change per mole of C_6H_{14} produced in this reaction.

enthalpy change = kJ mol^{-1} [2]

8. M/J 24/P21/Q4/a,b

4 (a) Define enthalpy change of formation.

[2]

(b) Iron is made when iron(III) oxide is heated with carbon monoxide, as shown by reaction 2.

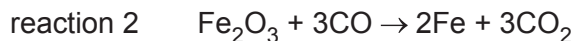


Table 4.1 shows enthalpy change of formation data measured at 298 K and 101 kPa.

Table 4.1

substance	equation	value for $\Delta H_f^\circ / \text{kJ mol}^{-1}$
Fe_2O_3		-824.2
CO		-110.5
CO_2	$\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$	-393.5

(i) Complete Table 4.1 by adding equations with relevant state symbols to represent:

- standard enthalpy change of formation for Fe_2O_3
- standard enthalpy change of formation for CO.

[2]

(ii) Use the data in Table 4.1 to calculate the enthalpy change of reaction, ΔH_r , in kJ mol^{-1} , for reaction 2.

Show your working.

$$\Delta H_r = \quad \quad \quad \text{kJ mol}^{-1} \quad [2]$$

9. M/J 24/P22/Q2/a,b

2 Separate samples of Na_2CO_3 and NaHCO_3 react with HCl(aq) to produce the same products, as shown in Table 2.1.

Table 2.1

reaction	equation	$\Delta H / \text{kJ mol}^{-1}$
1	$\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$	ΔH_1
2	$\text{NaHCO}_3 + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$	$\Delta H_2 = +27.2$

(a) Complete the reaction pathway diagram in Fig. 2.1 for reaction 2.

Label the diagram to show the enthalpy change, ΔH_2 , and the activation energy, E_A .

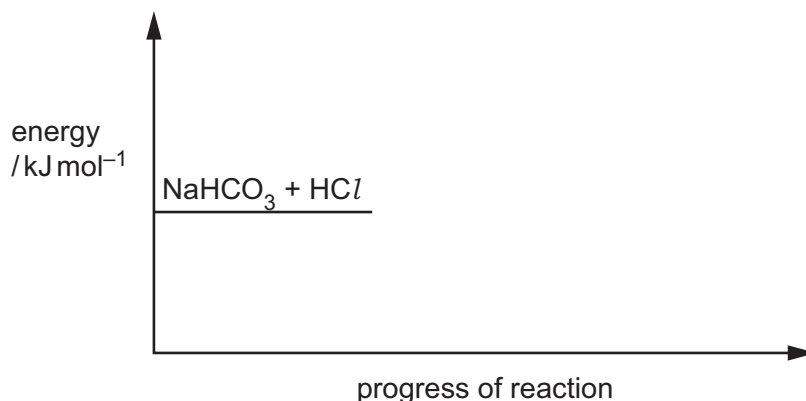


Fig. 2.1

[2]

(b) The value for ΔH_1 is determined by experiment using the following method.

- 50.0 cm³ of 2.00 mol dm⁻³ HCl(aq) is added to a polystyrene cup.
- The initial temperature of the acid is recorded as 19.6 °C.
- 0.0400 mol of Na₂CO₃ is added and the mixture is stirred.
- All the solid Na₂CO₃ disappears and a colourless solution is produced.

The maximum temperature recorded during the reaction is 26.2 °C.

(i) Describe **one** other observation that shows the reaction is complete.

[1]



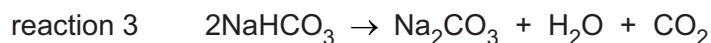
(ii) Calculate the value of ΔH_1 in kJ mol⁻¹.

Assume the specific heat capacity of the reaction mixture is the same as for water and no heat is lost to the surroundings.

Show your working.

$\Delta H_1 =$ kJ mol⁻¹ [3]

(iii) Thermal decomposition occurs when NaHCO₃ is heated.



Calculate the enthalpy change for reaction 3, ΔH_r , using the data in Table 2.1 and the value of ΔH_1 calculated in (b)(ii).

(If you were unable to calculate a value for ΔH_1 in (b)(ii), assume the enthalpy change is -38.4 kJ mol⁻¹. This is **not** the correct value.)

$\Delta H_r =$ kJ mol⁻¹ [2]

10. M/J 24/P23/Q3/a,b

- 3 (a)** Write an equation to show the reaction for the standard enthalpy change of formation of H_2O . Include state symbols.

[2]

- (b)** Water is one of the products in the reaction of B_2O_3 and NH_3 , as shown in reaction 2.

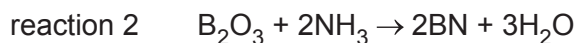


Table 3.1 shows information about the standard enthalpy change of formation, $\Delta H_f^\ominus$, of some substances.

Table 3.1

substance	$\Delta H_f^\ominus / \text{kJ mol}^{-1}$
B_2O_3	−1264
NH_3	−46
BN	−134
H_2O	−286

Calculate the enthalpy change, ΔH , for reaction 2 using the data from Table 3.1.

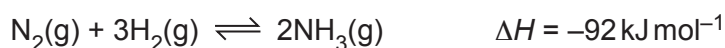
$\Delta H =$ kJ mol^{−1} [2]

11. MAR 24/P22/Q3/c(i)

- 3** Nitrogen, N_2 , is generally an unreactive molecule but it does react under certain conditions.

- (c)** The Haber process involves the reaction of N_2 and H_2 to form ammonia, NH_3 .

A catalyst is used, which allows the process to be carried out at a lower temperature and pressure.



- (i)** Use the information in **(c)** to complete Table 3.1.

Table 3.1

compound	enthalpy change of formation, $\Delta H_f^\ominus / \text{kJ mol}^{-1}$
N_2	
H_2	
NH_3	

[2]

12. O/N 23/P21/Q1/e

- 1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

- (e) POC_l₃(g) forms when PC_l₃(g) reacts with O₂(g).

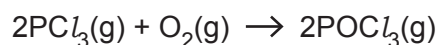


Table 1.3 gives some relevant data.

Table 1.3

process	value /kJ mol ⁻¹
enthalpy change of formation of PC _l ₃ (g)	-289
enthalpy change of formation of POC _l ₃ (g)	-592
O ₂ (g) → 2O(g)	+496

- (i) Define enthalpy change of formation, ΔH_f .

[2]

- (ii) Calculate the bond energy of P=O in POC_l₃ using the data in Table 1.3.

Show your working.

bond energy of P=O =

kJ mol⁻¹
[2]

13. O/N 23/P22/Q3/b(i,ii)

3 Phosphoric(V) acid, H_3PO_4 , is used in both inorganic and organic reactions.

(b) H_3PO_4 is also formed in the process shown in reaction 1.

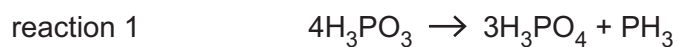


Table 3.1 shows some relevant thermodynamic data.

Table 3.1

compound	enthalpy change of formation, $\Delta H_f / \text{kJ mol}^{-1}$
H_3PO_3	-972
H_3PO_4	-1281
PH_3	+9

(i) Define enthalpy change of formation.

[2]

(ii) Use the data in Table 3.1 to calculate the enthalpy change, ΔH_r , of reaction 1.

$\Delta H_r =$ kJ mol^{-1}
[2]

14. O/N 23/P23/Q1/e

- 1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

- (e) POC_l₃(g) forms when PC_l₃(g) reacts with O₂(g).

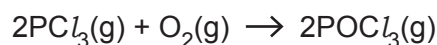


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O ₂ (g) → 2O(g)	+496

- (i) Define enthalpy change of formation, ΔH_f .

[2]

- (ii) Calculate the bond energy of P=O in POC_l₃ using the data in Table 1.3.

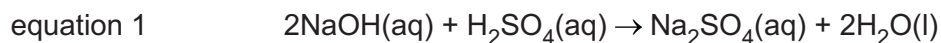
Show your working.

bond energy of P=O =

kJ mol⁻¹
[2]

15. M/J 23/P21/Q2/a,b(ii),c(ii)

2 A neutralisation reaction occurs when NaOH(aq) is added to H₂SO₄(aq).



(a) Define enthalpy change of neutralisation, ΔH_{neut} .

[2]

(b) An experiment is carried out to calculate ΔH_{neut} for the reaction between NaOH(aq) and H₂SO₄(aq).

100 cm³ of 1.00 mol dm⁻³ NaOH(aq) is added to 75 cm³ of 1.00 mol dm⁻³ H₂SO₄(aq) in a polystyrene cup and stirred. Results from the experiment are shown in Table 2.1.

Table 2.1

initial temperature of NaOH(aq)/°C	20.0
initial temperature of H ₂ SO ₄ (aq)/°C	20.0
maximum temperature of mixture/°C	27.8

(ii) Calculate ΔH_{neut} using the results in Table 2.1. Include units in your answer.

Assume that:

- the specific heat capacity of the final solution is 4.18 J g⁻¹ K⁻¹
- 1.00 cm³ of the final solution has a mass of 1.00 g
- there is no heat loss to the surroundings
- full dissociation of H₂SO₄(aq) occurs
- the experiment takes place at constant pressure.

Show your working.

$$\Delta H_{\text{neut}} = \quad \text{units}$$

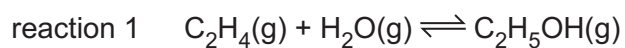
[3]

(ii) Suggest why the enthalpy change of neutralisation cannot be determined using the addition of dilute sulfuric acid to aqueous barium hydroxide.

[1]

16. M/J 23/P22/Q4/a

4 In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4 .



(a) Use the bond energy values in Table 4.1 to calculate the enthalpy change, ΔH_r , for reaction 1.

Table 4.1

bond	bond energy / kJ mol^{-1}
C–C	350
C=C	610
C≡C	840
C–H	410
C–O	360
C=O	740
O–H	460

$\Delta H_r =$ kJ mol^{-1} [2]

17. M/J 23/P23/Q3/a,b(ii),c(ii)

3 A neutralisation reaction occurs when NaOH(aq) is added to H₂SO₄(aq).



(a) Define enthalpy change of neutralisation, ΔH_{neut} .

[2]

(b) In an experiment, 50.0 cm³ of 2.00 mol dm⁻³ NaOH(aq) is added to 60.0 cm³ of 1.00 mol dm⁻³ H₂SO₄(aq) in a polystyrene cup and stirred. Both solutions have a temperature of 21.4 °C before mixing. The maximum temperature of the mixture is measured.

(ii) Calculate the theoretical maximum temperature of the mixture in this experiment.

Assume that:

- enthalpy change of neutralisation, ΔH_{neut} , of NaOH(aq) and H₂SO₄(aq) is -57.1 kJ mol⁻¹
- full dissociation of H₂SO₄(aq) occurs
- the specific heat capacity of the final solution is 4.18 J g⁻¹ K⁻¹
- 1.00 cm³ of the final solution has a mass of 1.00 g
- there is no heat loss to the surroundings
- the experiment takes place at constant pressure.

Show your working.

theoretical maximum temperature = °C [3]

(c) The enthalpy change of neutralisation of CH₃COOH(aq) and NaOH(aq) is -55.2 kJ mol⁻¹.

(ii) Values for the enthalpy change of neutralisation, ΔH_{neut} , are shown in Table 3.1.

Table 3.1

reagents	$\Delta H_{\text{neut}}/\text{kJ mol}^{-1}$
NaOH + HCl	-57.1
NaOH + CH ₃ COOH	-55.2

Suggest why the value for ΔH_{neut} of the weak acid, CH₃COOH, reacting with NaOH is different to the value obtained using the strong acid, HCl. Assume that the values are determined under the same conditions.

[1]

18. MAR 23/P22/Q3/d(iii)

- 3** Alkenes undergo an addition reaction with a 1:1 mixture of CO and H₂ to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H₂.

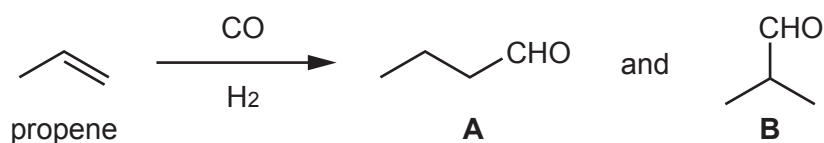
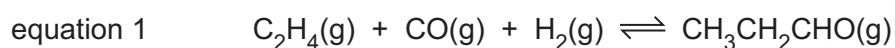


Fig. 3.1

- (d)** The reaction of ethene, C₂H₄, with a 1:1 mixture of CO and H₂ is shown in equation 1.



At atmospheric pressure a cobalt-based catalyst is used in this reaction.

- (iii)** Use the information in Table 3.2 to calculate the enthalpy change, ΔH_r , of the reaction in equation 1.

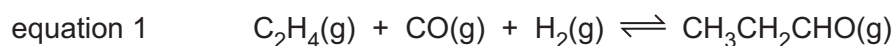


Table 3.2

compound	enthalpy change of formation, $\Delta H_f^\circ/\text{kJ mol}^{-1}$
C ₂ H ₄ (g)	+52
CO(g)	−111
CH ₃ CH ₂ CHO(g)	−187

$\Delta H_r =$ kJ mol^{−1} [2]

19. M/J 22/P21/Q1/f(ii)

- 1 Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.

(f) Magnesium, Mg, burns in oxygen, O₂.

The activation energy, E_a , for this reaction is +148 kJ mol⁻¹.

(ii) On Fig. 1.1:

- sketch a reaction pathway diagram for the reaction that occurs when Mg burns in O₂
- label the diagram to show the enthalpy change, ΔH , and the activation energy, E_a , for the reaction.

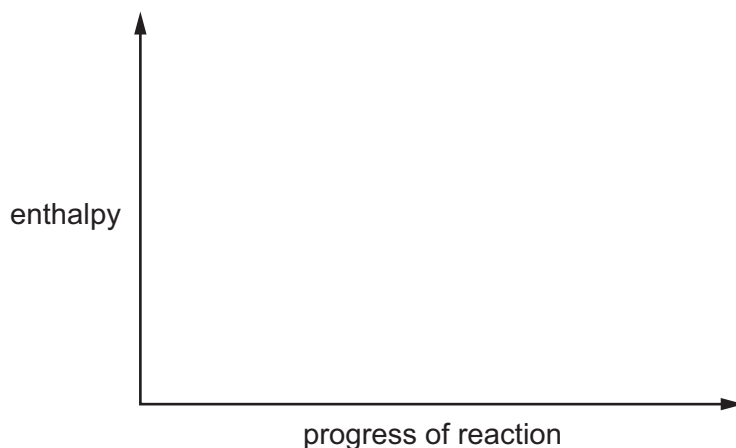


Fig. 1.1

[3]

20. M/J 22/P22/Q1/b(iii,iv)

- (b) When magnesium is heated in air, magnesium oxide, MgO, is the major product. Smaller amounts of magnesium nitride, Mg₃N₂, are also made.

(iii) Define enthalpy change of formation.

[2]

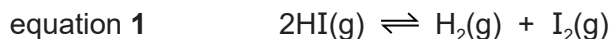
- (iv) When 3.645 g of Mg(s) burns in excess N₂(g) to form Mg₃N₂(s), 23.05 kJ of energy is released.

Calculate the enthalpy change of formation, ΔH_f° , of Mg₃N₂. Show your working.

$$\Delta H_f^\circ (\text{Mg}_3\text{N}_2) = \quad [3]$$

21. M/J 22/P22/Q3/b

- 3 (a)** 0.025 mol of HI(g) is added to a closed vessel and left to reach dynamic equilibrium. The total pressure of the vessel is 100 kPa.



- (b)** Use equation 1 and the bond energy values in Table 3.1 to calculate the change in enthalpy, ΔH , for the thermal decomposition of 1 mole of HI(g). Show your working.

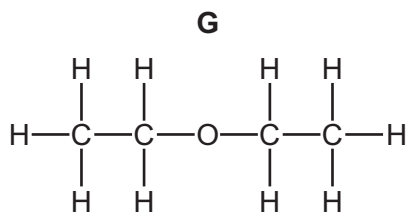
Table 3.1

bond	bond energy / kJ mol ⁻¹
H–H	436
I–I	151
H–I	299

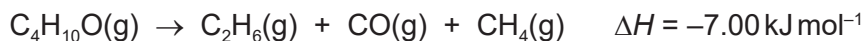
$$\Delta H = \quad \text{kJ mol}^{-1} \quad [2]$$

22. M/J 22/P23/Q3/a(ii)

- 3 G** belongs to a group of compounds called ethers.

**Fig. 3.1**

- (a)** When **G**, C₄H₁₀O, is heated, thermal decomposition occurs.



- (ii)** Calculate the bond energy of C≡O using the bond energy values in Table 3.1 and the enthalpy change, ΔH , for the thermal decomposition of **G**. Show your working.

Table 3.1

bond	bond energy / kJ mol ⁻¹
C–C	350
C–O (in G)	360
C–H	410

$$\text{bond energy (C}\equiv\text{O)} = \quad \text{kJ mol}^{-1} \quad [2]$$

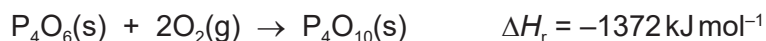
23. MAR 22/P22/Q2/c(iii)

2 Some oxides of elements in Period 3 are shown.



(c) P_4O_6 is a white solid that has a melting point of 24°C . Solid P_4O_6 reacts with water to form H_3PO_3 .

(iii) When $\text{P}_4\text{O}_6(\text{s})$ is heated with oxygen it forms $\text{P}_4\text{O}_{10}(\text{s})$.



The enthalpy change of formation, ΔH_{f} , of $\text{P}_4\text{O}_{10}(\text{s})$ is $-3012 \text{ kJ mol}^{-1}$.

Calculate the enthalpy change of formation, ΔH_{f} , of $\text{P}_4\text{O}_6(\text{s})$.

$$\Delta H_{\text{f}} \text{ of } \text{P}_4\text{O}_6(\text{s}) = \quad \quad \quad \text{kJ mol}^{-1} \quad [1]$$

24. O/N 21/P21/Q3/a(ii)

3 Phosphorus is a reactive Period 3 element.

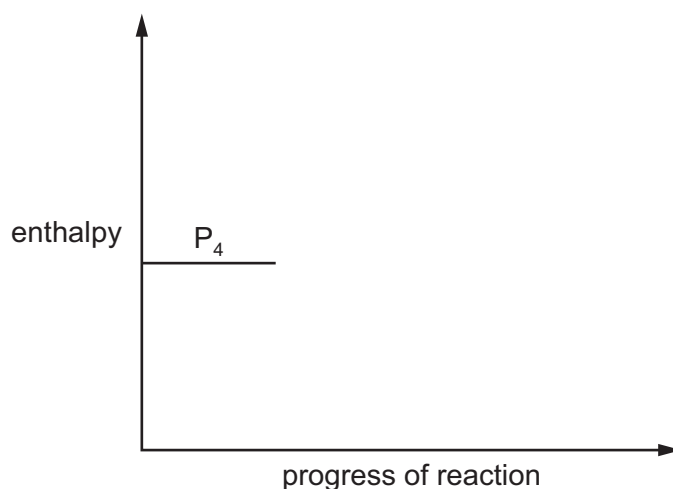
(a) Phosphorus has several allotropes. Details of two allotropes are given.

allotrope of phosphorus	formula	melting point/ $^\circ\text{C}$
white	P_4	44
red	P	590

(ii) Red phosphorus (P) forms when white phosphorus (P_4) is exposed to sunlight.



Use this information to draw a reaction pathway diagram to show the formation of red phosphorus (P) from white phosphorus (P_4).



[1]

25. O/N 21/P22/Q1/b

1 Hydrogen iodide, HI, is a colourless gas at room temperature.

(b) The standard enthalpy change of formation, $\Delta H_f^\ominus$, of HI(g) is +26.5 kJ mol⁻¹.

Define the term *standard enthalpy change of formation*

[2]

26. O/N 21/P23/Q3/a(ii)

3 Phosphorus is a reactive Period 3 element.

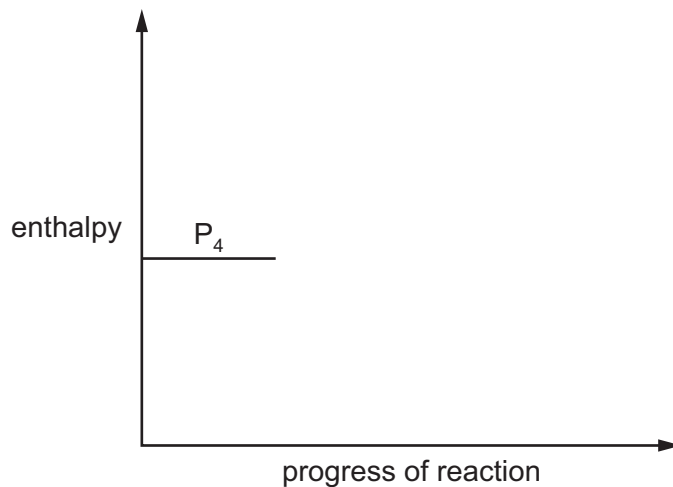
(a) Phosphorus has several allotropes. Details of two allotropes are given.

allotrope of phosphorus	formula	melting point/°C
white	P ₄	44
red	P	590

(ii) Red phosphorus (P) forms when white phosphorus (P₄) is exposed to sunlight.



Use this information to draw a reaction pathway diagram to show the formation of red phosphorus (P) from white phosphorus (P₄).

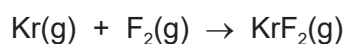


[1]

27. MAR 21/P22/Q1/c(i)

1 The rate of chemical reactions is affected by changes in temperature and pressure.

(c) Krypton reacts with fluorine in the presence of ultraviolet light to make krypton difluoride, $\text{KrF}_2(\text{g})$.

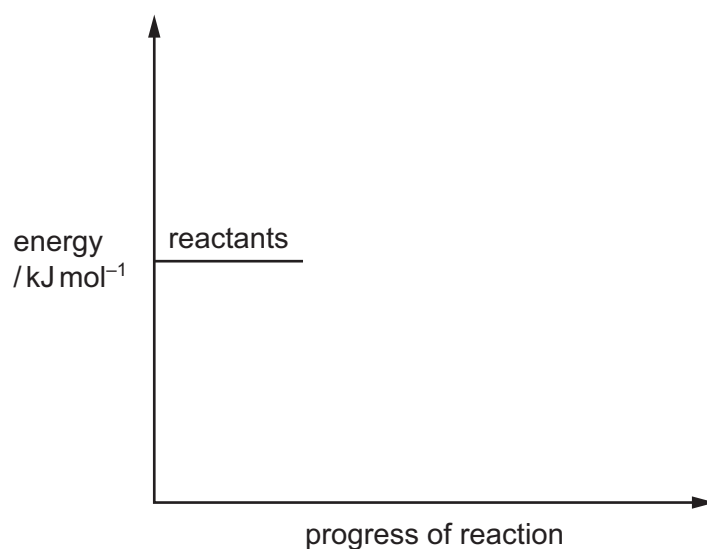


activation energy for the reaction, $E_a = +385 \text{ kJ mol}^{-1}$

enthalpy change of formation of KrF_2 , $\Delta H_f = +60.2 \text{ kJ mol}^{-1}$

(i) Use this information to complete the reaction profile diagram for the formation of KrF_2 . Label E_a and ΔH_f on the diagram.

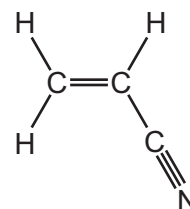
Assume the reaction proceeds in one step.



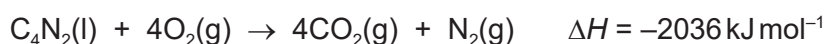
[2]

28. MAR 21/P22/Q3/a(i)

3 Compounds **P**, **Q** and **R** have all been found in the atmosphere of one of Saturn's moons.

P**Q****R**

(a) The equation for the complete combustion of **P**, $\text{C}_4\text{N}_2(\text{l})$, is shown.



(i) The enthalpy change of formation, ΔH_f , of $\text{CO}_2(\text{g})$ is -384 kJ mol^{-1} .

Calculate the enthalpy change of formation, ΔH_f , of **P**, in kJ mol^{-1} .

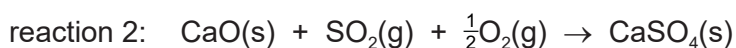
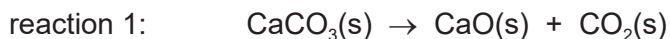
ΔH_f of **P** = kJ mol^{-1} [2]

29. O/N 20/P21/Q2/c(ii)

2 Phosphorus, sulfur and chlorine can all react with oxygen to form oxides.

- (c) Emissions of SO_2 from coal-fired power stations can be reduced by mixing the coal with powdered limestone.

Limestone is heated to form CaO in reaction 1. This then reacts with SO_2 and O_2 to form CaSO_4 in reaction 2.



- (ii) Use the data to calculate the enthalpy change of reaction 2.

compound	$\Delta H_f / \text{kJ mol}^{-1}$
CaO(s)	-635
SO ₂ (g)	-297
CaSO ₄ (s)	-1434

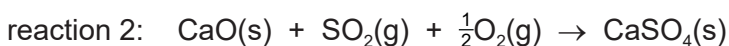
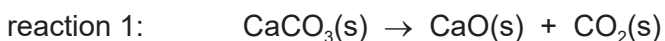
enthalpy change of reaction 2 = kJ mol⁻¹ [2]

30. O/N 20/P23/Q2/c(ii)

2 Phosphorus, sulfur and chlorine can all react with oxygen to form oxides.

- (c) Emissions of SO_2 from coal-fired power stations can be reduced by mixing the coal with powdered limestone.

Limestone is heated to form CaO in reaction 1. This then reacts with SO_2 and O_2 to form CaSO_4 in reaction 2.



- (ii) Use the data to calculate the enthalpy change of reaction 2.

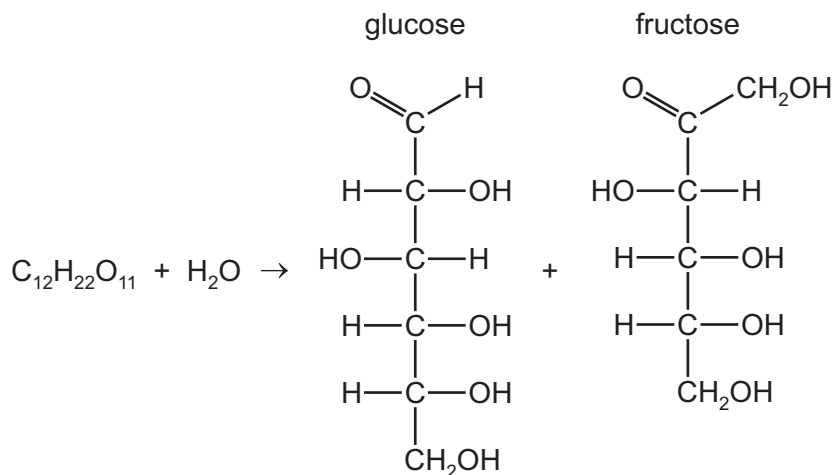
compound	$\Delta H_f / \text{kJ mol}^{-1}$
CaO(s)	-635
SO ₂ (g)	-297
CaSO ₄ (s)	-1434

enthalpy change of reaction 2 = kJ mol⁻¹ [2]

31. M/J 20/P22/Q3/d

Data Booklet needed

- 3** Sucrose, $C_{12}H_{22}O_{11}$, reacts with water to form glucose and fructose in reaction **A**.

reaction **A**

- (d)** 1.00 g of sucrose, $C_{12}H_{22}O_{11}$, is completely combusted. The heat energy produced is used to increase the temperature of 250 g of water inside a calorimeter from 25.0 °C to 40.7 °C.

These data can be used to calculate the enthalpy change of combustion of sucrose.

- (i)** Explain what is meant by the term *enthalpy change of combustion of sucrose*.

[2]

- (ii)** Use the *Data Booklet* to calculate the enthalpy change, in kJ mol^{-1} , for the combustion of sucrose.

Assume that all of the heat energy produced is transferred to the water.

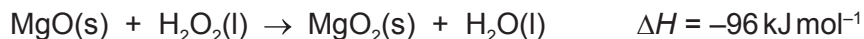
Show your working.

enthalpy change of combustion of sucrose = kJ mol^{-1}
[3]

32. MAR 20/P22/Q1/g(ii,iii,iv)

1 Group 2 metals form alkaline solutions in water.

(g) Magnesium peroxide, MgO_2 , is made in the following reaction.



compound	enthalpy change of formation, $\Delta H_f / \text{kJ mol}^{-1}$
MgO(s)	-602
H ₂ O ₂ (l)	-188
H ₂ O(l)	-286

(ii) Define the term *enthalpy change of formation*.

[2]

(iii) Use the data given to calculate the enthalpy change of formation of $\text{MgO}_2(\text{s})$.

$$\Delta H_f \text{ MgO}_2(\text{s}) = \quad \text{kJ mol}^{-1} \quad [2]$$

(iv) Magnesium peroxide decomposes slowly to form magnesium oxide and oxygen.



Use your answer to **(g)(iii)** and the data in the table to calculate the enthalpy change of this reaction.

*If you were unable to obtain an answer to **(g)(iii)**, use the value $\Delta H_f = -550 \text{ kJ mol}^{-1}$. This is **not** the correct answer.*

$$\text{enthalpy change of reaction} = \quad \text{kJ mol}^{-1} \quad [1]$$

33. O/N 19/P21/Q3/b(ii,iii,iv)

3 Crude oil is a natural source of hydrocarbons that are used as fuels.

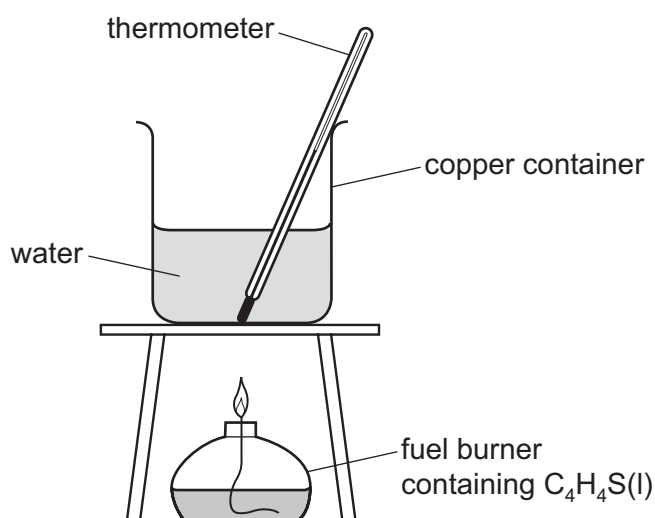
(b) Thiophene, $C_4H_4S(l)$, is an organic compound that is found as a contaminant in crude oil.

(ii) A student carries out an experiment to determine the enthalpy change of combustion of $C_4H_4S(l)$.

Explain the meaning of the term *enthalpy change of combustion*.

[2]

(iii) The student uses the following apparatus in the experiment.



mass of water in copper container / g	200
initial temperature of water / °C	18.5
highest temperature of water / °C	37.5

Calculate the heat energy released, in J, by the reaction.

Assume that 4.18 J of heat energy changes the temperature of 1.0 cm³ of water by 1.0 °C.

Assume no heat is lost to the surroundings.

heat energy released = _____ J
[2]

(iv) The student used 0.63 g of $C_4H_4S(l)$ in the experiment.

Calculate the enthalpy change of combustion of thiophene, $\Delta H_c(C_4H_4S(l))$. Include a sign in your answer.

$\Delta H_c(C_4H_4S(l)) =$ _____ kJ mol⁻¹
[2]

34. O/N 19/P23/Q3/b(ii,iii,iv)

3 Crude oil is a natural source of hydrocarbons that are used as fuels.

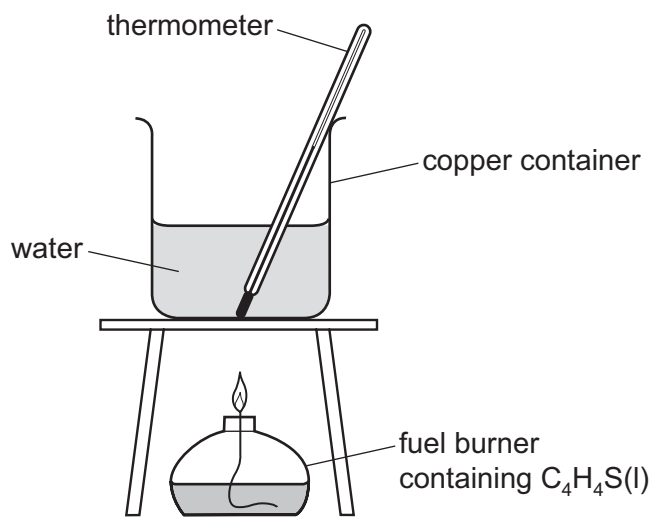
(b) Thiophene, $\text{C}_4\text{H}_4\text{S(l)}$, is an organic compound that is found as a contaminant in crude oil.

(ii) A student carries out an experiment to determine the enthalpy change of combustion of $\text{C}_4\text{H}_4\text{S(l)}$.

Explain the meaning of the term *enthalpy change of combustion*.

[2]

(iii) The student uses the following apparatus in the experiment.



mass of water in copper container / g	200
initial temperature of water / °C	18.5
highest temperature of water / °C	37.5

Calculate the heat energy released, in J, by the reaction.

Assume that 4.18 J of heat energy changes the temperature of 1.0 cm³ of water by 1.0 °C.

Assume no heat is lost to the surroundings.

heat energy released = _____ J
[2]

(iv) The student used 0.63 g of $\text{C}_4\text{H}_4\text{S(l)}$ in the experiment.

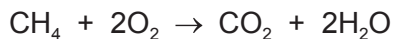
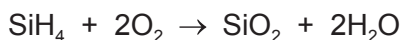
Calculate the enthalpy change of combustion of thiophene, $\Delta H_c(\text{C}_4\text{H}_4\text{S(l)})$. Include a sign in your answer.

$\Delta H_c(\text{C}_4\text{H}_4\text{S(l)}) =$ _____ kJ mol⁻¹
[2]

35. M/J 19/P21/Q2/e

Data Booklet needed

- 2 Magnesium silicide, Mg_2Si , is a compound made by heating magnesium with sand.
- (e) SiH_4 reacts in air without heating but CH_4 must be ignited before combustion occurs.



Suggest, with reference to bond energies from the *Data Booklet*, why SiH_4 reacts in air without heating but CH_4 must be ignited.

[2]

36. MAR 19/P22/Q2/c(iii)

- 2 The elements in Group 17 of the Periodic Table are called the halogens. They form stable compounds with both metals and non-metals.

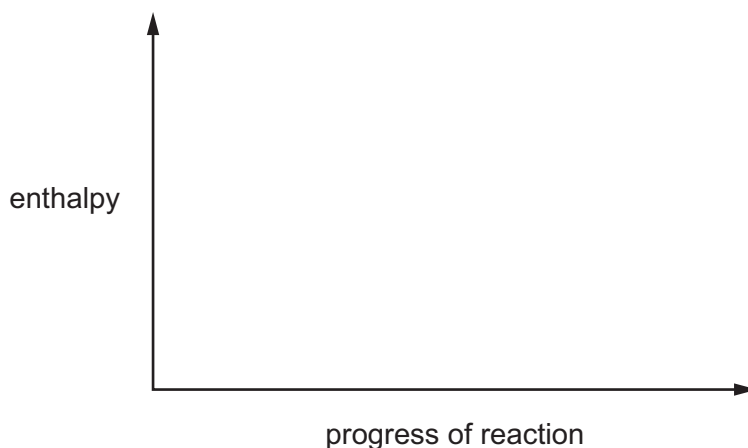
The table gives some data about F_2 , HCl and CaF_2 .

	F_2	HCl	CaF_2
boiling point/K	85	188	2773
relative formula mass	38.0	36.5	78.1

- (c) HOF is the only known molecule that contains only the elements hydrogen, oxygen and fluorine.
- (iii) HOF is an unstable compound and decomposes to form HF and O_2 .



Draw a fully labelled reaction pathway diagram on the axes provided to show the decomposition of HOF into HF and O_2 .



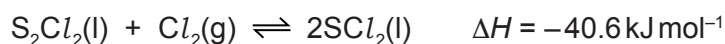
[2]

37. O/N 18/P22/Q2/b(ii)

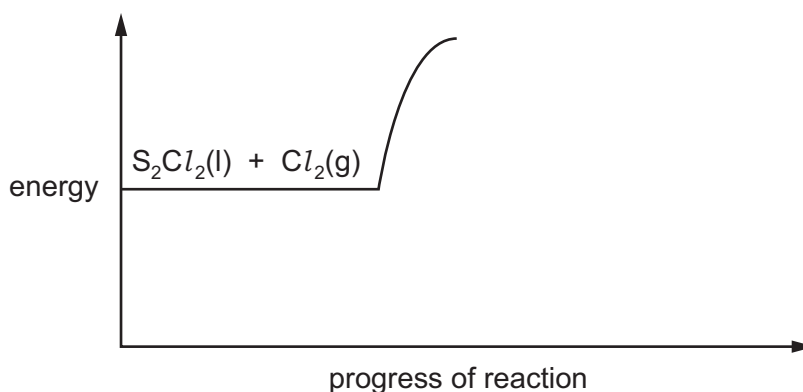
2 The table gives some data for elements in the third period and some of their compounds.

element	Na	Mg	Al	Si	P	S
type of bonding	metallic				covalent	covalent
formula of oxide					P ₄ O ₁₀	SO ₂
formula of chloride	NaCl	MgCl ₂				SCl ₂

(b) SCl₂ is formed in the following reaction.



(ii) Complete and fully label the reaction pathway diagram for the reaction between S₂Cl₂ and Cl₂. Include labels for activation energy, E_a , and enthalpy change of the forward reaction, ΔH .

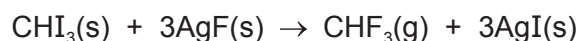


[2]

38. O/N 18/P22/Q3/a

3 Trihalomethanes are organic molecules in which three of the hydrogen atoms of methane are replaced by halogen atoms, for example CHF₃.

(a) The equation shows a reaction to produce CHF₃.



Use the data to calculate the enthalpy change of reaction, ΔH_r , for this formation of CHF₃.

compound	enthalpy change of formation, $\Delta H_f^\circ / \text{kJ mol}^{-1}$
CHI ₃ (s)	-182.1
CHF ₃ (g)	-692.9
AgF(s)	-204.6
AgI(s)	-61.8

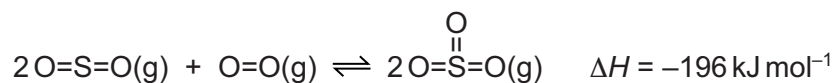
enthalpy change of reaction, $\Delta H_r =$

kJ mol⁻¹ [3]

39. M/J 18/P21/Q1/b

- 1 Sulfuric acid is manufactured by the Contact process.

One stage in this process is the conversion of sulfur dioxide into sulfur trioxide in the presence of a heterogeneous catalyst of vanadium(V) oxide, V_2O_5 .



- (b) Some bond energies are given.

bond	bond energy / kJ mol^{-1}
S=O (in SO_2)	534
O=O	496

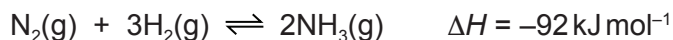
Use the data, and the enthalpy change for the conversion of sulfur dioxide into sulfur trioxide, to calculate a value for the S=O bond energy in SO_3 .

$$\text{S=O bond energy in } \text{SO}_3 = \quad \quad \quad \text{kJ mol}^{-1} \quad [2]$$

The Contact process is usually carried out at a temperature of about 400°C and a pressure just above atmospheric pressure. Using a higher or lower temperature and pressure would affect both the rate of production of sulfur trioxide and the yield of sulfur trioxide.

40. O/N 17/P21/Q1/a

- 1 Ammonia, NH_3 , is manufactured from nitrogen and hydrogen by the Haber process.



- (a) Some bond energies are given.

$$\text{N}\equiv\text{N} = 944 \text{ kJ mol}^{-1}$$

$$\text{H}-\text{H} = 436 \text{ kJ mol}^{-1}$$

- (i) Explain the meaning of the term *bond energy*.

[2]

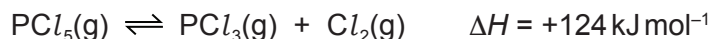
- (ii) Use the data to calculate a value for the N–H bond energy.

You must show your working.

$$\text{N}-\text{H bond energy} = \quad \quad \quad \text{kJ mol}^{-1} \quad [2]$$

41. O/N 17/P22/Q2/a

- 2 At 450 K phosphorus(V) chloride, $\text{PCl}_5(\text{g})$, decomposes to form phosphorus(III) chloride, $\text{PCl}_3(\text{g})$, and chlorine, $\text{Cl}_2(\text{g})$. A dynamic equilibrium is established as shown.



- (a) The enthalpy change of formation of $\text{PCl}_3(\text{g})$ under these conditions is given.

$$\Delta H_f \text{ PCl}_3(\text{g}) = -320 \text{ kJ mol}^{-1}$$

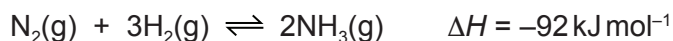
Calculate the enthalpy change of formation of $\text{PCl}_5(\text{g})$ under these conditions.

Include a sign with your answer.

enthalpy change = kJ mol⁻¹ [1]

42. O/N 17/P23/Q1/a

- 1 Ammonia, NH_3 , is manufactured from nitrogen and hydrogen by the Haber process.



- (a) Some bond energies are given.

$$\text{N} \equiv \text{N} = 944 \text{ kJ mol}^{-1}$$

$$\text{H}-\text{H} = 436 \text{ kJ mol}^{-1}$$

- (i) Explain the meaning of the term *bond energy*.

[2]

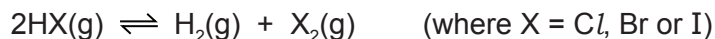
- (ii) Use the data to calculate a value for the N–H bond energy.

You must show your working.

N–H bond energy = kJ mol⁻¹ [2]

43. M/J 17/P22/Q3/a

- 3 The hydrogen halides, HCl , HBr and HI , can undergo thermal decomposition. In a sealed container an equilibrium is established according to the equation shown.



- (a) Some bond energies are shown in the table.

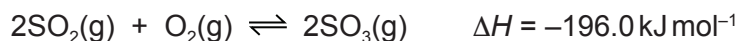
	bond energy / kJ mol ⁻¹
H–Br	366
H–H	436
Br–Br	193

Use these data to calculate a value for the enthalpy change, ΔH , for the thermal decomposition of hydrogen bromide, HBr , according to the equation shown.

$\Delta H =$ kJ mol⁻¹ [1]

44. M/J 17/P23/Q3/a

- 3 Sulfur trioxide, SO_3 , is manufactured from sulfur dioxide and oxygen by the Contact process.



- (a) The enthalpy change of formation of SO_2 , $\Delta H_f^\circ \text{SO}_2(\text{g})$, is $-296.8 \text{ kJ mol}^{-1}$.

- (i) Define the term *enthalpy change of formation*.

[2]

- (ii) Use the data to calculate the enthalpy change of formation of $\text{SO}_3(\text{g})$.

$$\Delta H_f^\circ \text{SO}_3(\text{g}) = \quad \text{kJ mol}^{-1} \quad [2]$$

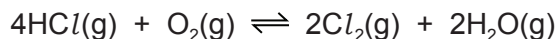
45. MAR 17/P22/Q2/c(i)

- 2 Hydrogen halides are compounds formed when halogens (Group 17 elements) react with hydrogen. The bond polarity of the hydrogen halides decreases from HF to HI.

Some relevant data are shown in the table.

hydrogen halide	HF	HCl	HBr	HI
boiling point/ $^\circ\text{C}$	19	-85	-67	-35
H-X bond energy/ kJ mol^{-1}	562	431	366	299

- (c) Hydrogen chloride undergoes a reversible reaction with oxygen.



The reaction is carried out at 400°C in the presence of a copper(II) chloride catalyst.

- (i) Use the data in the table to calculate the overall enthalpy change of reaction.

compound	enthalpy change of formation/ kJ mol^{-1}
$\text{HCl}(\text{g})$	-92
$\text{H}_2\text{O}(\text{g})$	-242

$$\text{enthalpy change of reaction} = \quad \text{kJ mol}^{-1} \quad [2]$$

46. O/N 16/P21/Q2/a,b,c

- 2** For many compounds the enthalpy change of formation cannot be calculated directly. An indirect method based on enthalpy changes of combustion can be used.

The enthalpy change of combustion can be found by a calorimetry experiment in which the heat energy given off during combustion is used to heat a known mass of water and the temperature change recorded.

- (a) (i)** Explain the meaning of the term *standard enthalpy change of combustion*.

[3]

- (ii)** Write the equation for the complete combustion of ethanol, C₂H₅OH.

[1]

- (b)** In an experiment to determine the enthalpy change of combustion of ethanol, 0.23 g of ethanol was burned and the heat given off raised the temperature of 100 g of water by 16.3 °C.

- (i)** Calculate the heat energy change, q , during the combustion of 0.23 g of ethanol.

$$q = \quad \text{J} \quad [1]$$

- (ii)** Calculate the enthalpy change on burning 1 mole of ethanol. Include a sign in your answer.

$$\Delta H = \quad \text{kJ mol}^{-1} \quad [1]$$

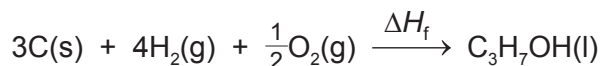
- (iii)** Suggest **two** reasons why the value for the enthalpy change of combustion of ethanol determined by a simple laboratory calorimetry experiment is likely to be lower than the true value.

[2]

- (c)** The table gives some enthalpy change of combustion values.

substance	enthalpy change of combustion / kJ mol ⁻¹
C(s)	-393.5
H ₂ (g)	-285.8
C ₃ H ₇ OH(l)	-2021.0

- (i)** Construct a labelled energy cycle to show how these values could be used to calculate the enthalpy change of formation of C₃H₇OH(l), ΔH_f .



[3]

- (ii)** Calculate the enthalpy change of formation, ΔH_f , of C₃H₇OH(l).

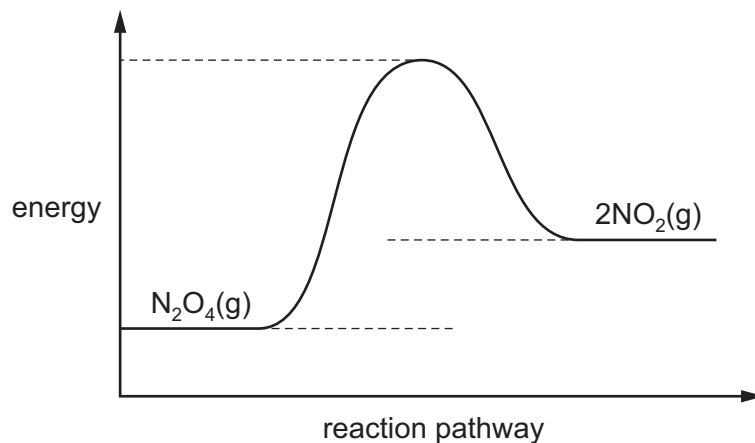
$$\Delta H_f = \quad \text{kJ mol}^{-1} \quad [2]$$

47. O/N 16/P22/Q2/a

2 Dinitrogen tetraoxide, N_2O_4 , and nitrogen dioxide, NO_2 , exist in dynamic equilibrium with each other.



The energy profile for this reaction is shown.



(a) Add labelled arrows to the energy profile to indicate

- the enthalpy change of the reaction, ΔH ,
- the activation energy of the forward reaction, E_a .

[2]

48. O/N 16/P23/Q2/a,b,c

- 2 For many compounds the enthalpy change of formation cannot be calculated directly. An indirect method based on enthalpy changes of combustion can be used.

The enthalpy change of combustion can be found by a calorimetry experiment in which the heat energy given off during combustion is used to heat a known mass of water and the temperature change recorded.

- (a) (i) Explain the meaning of the term *standard enthalpy change of combustion*.

[3]

- (ii) Write the equation for the complete combustion of ethanol, C₂H₅OH.

[1]

- (b) In an experiment to determine the enthalpy change of combustion of ethanol, 0.23 g of ethanol was burned and the heat given off raised the temperature of 100 g of water by 16.3 °C.

- (i) Calculate the heat energy change, q , during the combustion of 0.23 g of ethanol.

$$q = \quad \text{J} \quad [1]$$

- (ii) Calculate the enthalpy change on burning 1 mole of ethanol. Include a sign in your answer.

$$\Delta H = \quad \text{kJ mol}^{-1} \quad [1]$$

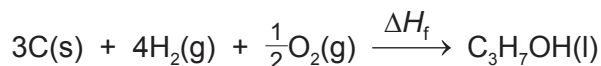
- (iii) Suggest **two** reasons why the value for the enthalpy change of combustion of ethanol determined by a simple laboratory calorimetry experiment is likely to be lower than the true value.

[2]

- (c) The table gives some enthalpy change of combustion values.

substance	enthalpy change of combustion / kJ mol ⁻¹
C(s)	-393.5
H ₂ (g)	-285.8
C ₃ H ₇ OH(l)	-2021.0

- (i) Construct a labelled energy cycle to show how these values could be used to calculate the enthalpy change of formation of C₃H₇OH(l), ΔH_f .



[3]

- (ii) Calculate the enthalpy change of formation, ΔH_f , of C₃H₇OH(l).

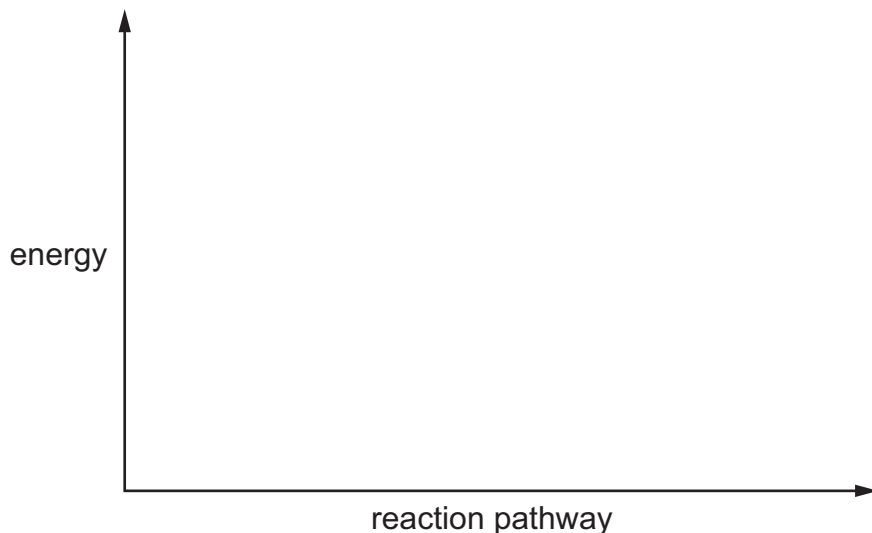
$$\Delta H_f = \quad \text{kJ mol}^{-1} \quad [2]$$

49. M/J 16/P21/Q3/b(i)

3 The elements in Group 2, and their compounds, show many similarities and trends in their properties.

(b) Magnesium oxide can be formed by the reaction of magnesium and oxygen in the air.

(i) Draw a **fully labelled** reaction pathway diagram for the reaction between magnesium and oxygen.



[2]

50. M/J 16/P22/Q2/a(iii)

2 The elements in Group 17, the halogens, and their compounds, show many similarities and trends in their properties. Some data are given for the elements fluorine to iodine.

element	bond energy / kJ mol^{-1}	standard enthalpy change of atomisation, $\Delta H_{\text{at}}^{\ominus}$ / kJ mol^{-1}	boiling point of element / K	boiling point of hydrogen halide / K
fluorine, F–F	158	79	85	293
chlorine, Cl–Cl	242	121	238	188
bromine, Br–Br	193	112	332	206
iodine, I–I	151	107	457	238

(iii) The standard enthalpy of formation of iodine monochloride, ICl , is $-24.0 \text{ kJ mol}^{-1}$.

Use this information and the bond energies of iodine and chlorine to calculate the I–Cl bond energy.

I–Cl bond energy = kJ mol^{-1} [2]

51. O/N 15/P21/Q2/a,b(i)

Data Booklet needed

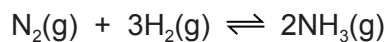
2 (a) (i) Explain the meaning of the term *enthalpy change of formation*.

[2]

(ii) Give the equation for the reaction for which the enthalpy change corresponds to the standard enthalpy change of formation of liquid sulfur trioxide, SO₃. Include state symbols.

[1]

(b) Ammonia is manufactured by the Haber process.



(i) Use bond energies from the *Data Booklet* to calculate the enthalpy change of reaction for the Haber process. Include a sign in your answer.

enthalpy change

kJ mol⁻¹ [3]

52. O/N 15/P22/Q2/a,b

2 Chemical reactions are accompanied by enthalpy changes.

(a) Explain the meaning of the term *standard enthalpy change* of reaction.

[2]

(b) The enthalpy change of hydration of anhydrous magnesium sulfate, ΔH_{hyd} MgSO_4 , can be calculated by carrying out two separate experiments.

In the first experiment 45.00g of water was weighed into a polystyrene cup and 3.01g of MgSO_4 was added and stirred until it was completely dissolved. The temperature of the water rose from 23.4 °C to 34.7 °C.

(i) Calculate the amount of heat energy transferred to the water during this dissolving process.

You can assume that the specific heat capacity of the solution is the same as that of water, $4.18 \text{ J g}^{-1} \text{ K}^{-1}$.

heat energy = J [1]

(ii) Calculate the amount, in moles, of MgSO_4 dissolved.

amount = mol [1]

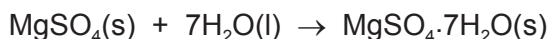
(iii) Calculate the enthalpy change of solution, ΔH_{soln} , of $\text{MgSO}_4(\text{s})$.

You must include a sign with your answer.

ΔH_{soln} , of $\text{MgSO}_4(\text{s})$ = kJ mol^{-1} [1]

In the second experiment, the enthalpy change of solution for the hydrated salt, $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}(\text{s})$, was calculated and found to be $+9.60 \text{ kJ mol}^{-1}$.

(iv) Use the equation below for the hydration of anhydrous magnesium sulfate to construct a suitable, fully labelled energy cycle that will allow you to calculate the enthalpy change for this reaction, ΔH_{hyd} MgSO_4 .



[1]

(v) Calculate the enthalpy change for this reaction, ΔH_{hyd} MgSO_4 . Include a sign in your answer.

ΔH_{hyd} MgSO_4 = kJ mol^{-1} [1]

53. O/N 15/P23/Q2/a,b(i)

Data Booklet needed

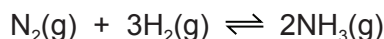
- 2 (a) (i)** Explain the meaning of the term *enthalpy change of formation*.

[2]

- (ii)** Give the equation for the reaction for which the enthalpy change corresponds to the standard enthalpy change of formation of liquid sulfur trioxide, SO₃. Include state symbols.

[1]

- (b)** Ammonia is manufactured by the Haber process.



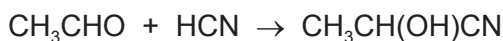
- (i)** Use bond energies from the *Data Booklet* to calculate the enthalpy change of reaction for the Haber process. Include a sign in your answer.

enthalpy change kJ mol⁻¹ [3]

54. M/J 15/P21/Q3/a

Data Booklet needed

- 3** Ethanal reacts with hydrogen cyanide, in the presence of a small amount of NaCN, as shown.



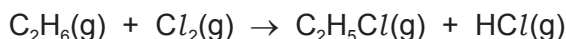
- (a)** Use bond energies from the *Data Booklet* to calculate the enthalpy change for this reaction. Include a sign with your answer.

enthalpy change = kJ mol⁻¹ [3]

55. M/J 15/P22/Q3/a(i)

Data Booklet needed

- 3** Ethane reacts with chlorine to form chloroethane.

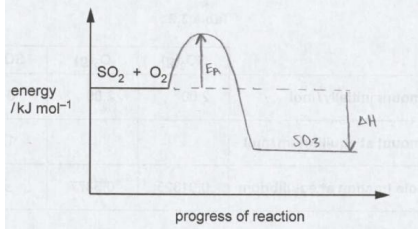


- (a) (i)** Use bond energies from the *Data Booklet* to calculate the enthalpy change for this reaction. Include a sign in your answer.

enthalpy change = kJ mol⁻¹ [3]

Answers Section

1. M/J 26/P23/Q3/a

3(a)	 - correct shape profile - arrow label ΔH from reactant line to product line - arrow label E_a (from reactant line to maximum on hump)	2 max 1 endothermic profile with correct labels & arrows for E_a and / or ΔH ΔH single headed arrows used in correct direction only. A E_a shown as a double headed arrow BOD vertical line A half-headed arrows A 'square' humps I missing products / incorrect labels on product line I '-' sign on $-\Delta H$ BOD poorly represented lines / arrows if approximate position and length (and direction) is correct
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2. MAR 26/P22/Q2/d(i)

2(d)(i)	- reactants higher than labelled products AND hump in the middle - reactants to top of hump = E_a OR (+)570 (kJ mol⁻¹) - reactants to products = ΔH OR (-)373 (kJ mol⁻¹)	2
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3. O/N 25/P21/Q3/d(ii)

3(d)(ii)	M1 (reaction 1): $-3115 - 286 + 3292 = -109$ (kJ mol⁻¹) M2 $\Delta H_2 = -53 - (-109) = (+)56$ (kJ mol⁻¹)	2
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4. O/N 25/P22/Q3/e(ii)

3(e)(ii)	M1 $+215 + (-156) - (+48) =$ M2 $+11$ kJ mol⁻¹	2
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5. O/N 25/P23/Q3/d(ii)

3(d)(ii)	M1 (reaction 1): $-3115 - 286 + 3292 = -109$ (kJ mol⁻¹) M2 $\Delta H_2 = -53 - (-109) = (+)56$ (kJ mol⁻¹)	2
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6. O/N 25/P24/Q1/b(iii,iv)

1(b)(iii)	M1 (enthalpy change / energy change) when one mole of a compound / substance is formed M2 from its (constituent) elements in their standard state(s)	2
1(b)(iv)	M1 $(6 \times -92) - (2 \times -46) =$ M2 -460 (kJ mol⁻¹)	2

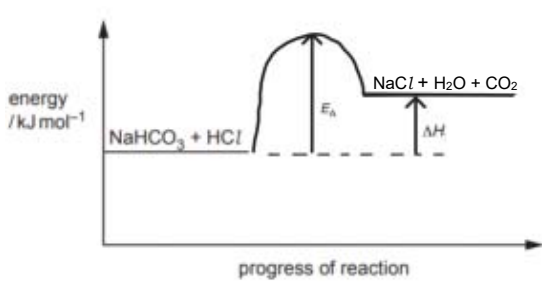
7. M/J 25/P22/Q3/c

3(c)	method 1 – breaking all bonds in molecules $(2 \times 610) + (2 \times 436) + (10 \times 410) + (3 \times 350) - (14 \times 410) - (5 \times 350)$ - energy involved when all bonds are broken = (7242) - energy involved when all bonds are made = (7490) - energy to break bonds – energy to make bonds $= (7242 - 7490) = -248$ method 2 – ignore any bonds that do not change from reactants to products $(2 \times 610) + (2 \times 436) - (4 \times 410) - (2 \times 350)$ - energy involved when all bonds are broken = (2092) - energy involved when all bonds are made = (2340) - energy to break bonds – energy to make bonds $(2092 - 2340) = -248$	2
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8. M/J 24/P21/Q4/a,b

4(a)	M1 (enthalpy change / energy change) when one mole of a compound / substance is formed M2 from its elements in their standard states	2												
4(b)(i)	<table border="1"> <thead> <tr> <th></th><th>equation for reaction</th><th>$\Delta H_{\text{formation}}$</th></tr> </thead> <tbody> <tr> <td>Fe_2O_3</td><td>$2\text{Fe (s)} + \frac{3}{2}\text{O}_2\text{(g)} \rightarrow \text{Fe}_2\text{O}_3\text{(s)}$</td><td>-824.2</td></tr> <tr> <td>CO</td><td>$\text{C (s)} + \frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{CO (g)}$</td><td>-110.5</td></tr> <tr> <td>CO_2</td><td>$\text{C (s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$</td><td>-393.5</td></tr> </tbody> </table>		equation for reaction	$\Delta H_{\text{formation}}$	Fe_2O_3	$2\text{Fe (s)} + \frac{3}{2}\text{O}_2\text{(g)} \rightarrow \text{Fe}_2\text{O}_3\text{(s)}$	-824.2	CO	$\text{C (s)} + \frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{CO (g)}$	-110.5	CO_2	$\text{C (s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$	-393.5	2
	equation for reaction	$\Delta H_{\text{formation}}$												
Fe_2O_3	$2\text{Fe (s)} + \frac{3}{2}\text{O}_2\text{(g)} \rightarrow \text{Fe}_2\text{O}_3\text{(s)}$	-824.2												
CO	$\text{C (s)} + \frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{CO (g)}$	-110.5												
CO_2	$\text{C (s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$	-393.5												
4(b)(ii)	$\Delta H_{\text{reaction}} = -\Delta H_{\text{Fe}_2\text{O}_3} - (3 \times -\Delta H_{\text{CO}}) + (3 \times -\Delta H_{\text{CO}_2})$ $= -(-824.2) - (3 \times -110.5) + (3 \times -393.5)$ $= -24.8 \text{ kJ mol}^{-1}$ - evidence of use of correct expression and / or any rearrangement of only these bond energy values - use correct stoichiometry for these bond energy values only - calculate their expression correctly provided the expression uses only addition / subtraction and all 3 values	2												

9. M/J 24/P22/Q2/a,b

2(a)	 <i>All three correct (2 marks) or two correct (1 mark):</i> - profile for endothermic reaction (include 'hump') - arrow representing E_a - arrow representing ΔH OR (+) 27.2	2
2(b)(i)	no more fizzing / effervescence stops	1

2(b)(ii)	M1 calculate Q from $= 50 \times 4.18 \times (26.2 - 19.6) = 1379.4$ M2 Use M1 to find energy change when 1 mol Na_2CO_3 / 2 mol HCl reacts M1 $\div 0.04 = 34485$ M3 express Q as $\Delta H_{(1)}$ in $\text{kJ mol}^{-1} = -(\text{value for M2}) \div 1000$ $= -34.5 \text{ kJ mol}^{-1}$ Alternative method M1 evidence of use $Q = mc\Delta T = 50 \times 4.18 / 1000 \times (26.2 - 19.6) = 1.3794$ M2 Use M1 to find energy change when 1 mol Na_2CO_3 / 2 mol HCl reacts M1 $\div 0.04 = 34.485$ M3 express Q as $\Delta H_{(1)}$ in $\text{kJ mol}^{-1} = -(\text{value for M2})$ (expressed in kJ) $= -34.5 \text{ kJ mol}^{-1}$	3
2(b)(iii)	Using answer from 2(b)(ii) $(=) 2\Delta H_2 - (\Delta H_1)$ $(=) 2 \times 27.2 - (2b(ii))$ $2 \times 27.2 - (-34.50) = (+) 88.9 \text{ kJ mol}^{-1}$ All three correct (2 marks) or two correct (1 mark): $(27.2) - (-34.5)$ OR $(27.2) - (2b(ii))$ 2×27.2 and $1 \times 2b(ii)$ - calculate correctly their expression Using alternative value of $\Delta H_1 = -38.4 \text{ kJ mol}^{-1}$ $\Delta H_r = (+)92.8 \text{ kJ mol}^{-1} = 2$ marks	2

10. M/J 24/P23/Q3/a,b

3(a)	$\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l})$ M1 correct equation only M2 correct state symbols based on this equation	2
3(b)	$\Delta H = -(-1264) - (2 \times -46) + (2 \times -134) + (3 \times -286) = (+)230 \text{ kJ mol}^{-1}$ M1 correct expression in terms of all 4 enthalpy changes of formation AND correct stoichiometry (as described in given equation) M2 correct calculation based on figures given	2

11. MAR 24/P22/Q3/c(i)

3(c)(i)	0 for both ΔH_f of N_2 and H_2	1
	-46 for ΔH_f of NH_3	1

12. O/N 23/P21/Q1/e

1(e)(i)	enthalpy change when one mole of a compound / substance is formed	1
	from its constituent elements in their standard states	1
1(e)(ii)	$2(-592) = 2(-289) + (496) - 2(\text{P}=\text{O})$ OR $(2) \text{P}=\text{O} = 2(-289) + (496) + 2(592)$	1
	$E_{\text{P}=\text{O}} = \frac{1}{2}(1102) = +551 (\text{kJ mol}^{-1})$	1

13. O/N 23/P22/Q3/b(i,ii)

3(b)(i)	enthalpy change when one mole of a compound / substance is formed	1
	from its constituent elements in their standard states	1
3(b)(ii)	$\Delta H_f = +9 + 3(-1281) - 4(-972) = (+)54 (\text{kJ mol}^{-1}) (1)$ Any two from (1): - use correct stoichiometry for all three values - use correct expression regardless of stoichiometry for all three values - calculated correctly	2

14. O/N 23/P23/Q1/e

1(e)(i)	enthalpy change when one mole of a compound / substance is formed	1
	from its constituent elements in their standard states	1
1(e)(ii)	$2(-592) = 2(-289) + (496) - 2(P=O)$ OR $(2) P=O = 2(-289) + (496) + 2(592)$	1
	$E_{P=O} = \frac{1}{2}(1102) = +551 \text{ (kJ mol}^{-1}\text{)}$	1

15. M/J 23/P21/Q2/a,b(ii),c(ii)

2(a)	M1 (enthalpy change) when 1 mole of water is formed M2 from an (aqueous) acid and an alkali / (aqueous) base	2
2(b)(ii)	M1 correct volume and correct temperature change M2 calculation to calculate Q based on M1 volume and temp and $Q = mc\Delta T$ $Q = mc\Delta T = 175 \times 4.18 \times (27.8 - 20) = (-)5705.7$ M3 energy change when 0.1 mol H_2O is made with appropriate unit and sign $\Delta H = M2 / 0.1 = -57.1 \text{ kJ mol}^{-1}$ OR $-57100 \text{ J mol}^{-1}$	3
2(c)(ii)	precipitation (of $BaSO_4$) (also) involves an energy change	1

16. M/J 23/P22/Q4/a

4(a)	Method 1 - identification of number and type of bonds when all are broken $2O-H + 4C-H + C=C$ OR $(2 \times 460) + (4 \times 410) + 610$ - Identification of number and type of bonds when all are made $O-H + 5C-H + C-C + C-O$ OR $(460 + (5 \times 410) + 350 + 360)$ - calculation of energy to break bonds – energy to make bonds $3170 - 3220 = -50 \text{ (kJ mol}^{-1}\text{)}$ Two correct for one mark, three correct for two marks. OR Method 2 - identification of number and type of relevant bonds broken (including the new C-H only) $2O-H + C=C$ OR $(2 \times 460) + 610$ - identification of number and type of relevant bonds made (including the new C-H only) $O-H + C-H + C-C + C-O$ OR $(460 + 410 + 350 + 360)$ - calculation using energy to break bonds – energy to make bonds $1530 - 1580 = -50 \text{ (kJ mol}^{-1}\text{)}$ Two correct for one mark, three correct for two marks.	2
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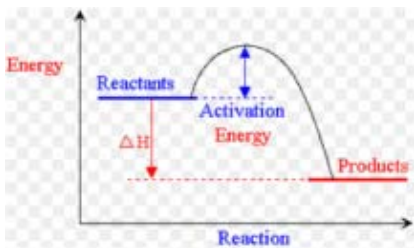
17. M/J 23/P23/Q3/a,b(ii),c(ii)

3(a)	M1 (energy change) when 1 mole of water is formed M2 from an (aqueous) acid and an alkali / aqueous base	2
3(b)(ii)	M1 correct value for Q (using 0.1 mol H_2O / OR $2 \times (b)(i)$) (and Q in Joules) M2 calculate ΔT (using their Q AND 110) $Q/mc = \Delta T = (Q) \div (4.18 \times 110) = 12.42 \text{ (}^\circ\text{C)}$ M3 use ΔT and initial temperature to find temperature of final solution $21.4 + M2 = 33.8 \text{ (}^\circ\text{C)}$ [M3 $\Delta T + 21.4 = T_{\text{max}}$]	3
3(c)(ii)	(additional) energy is involved / required to fully ionise / dissociate (weak / ethanoic acid) OR (additional) energy is involved / required to donate/ release H^+ ions	1

18. MAR 23/P22/Q3/d(iii)

3(d)(iii)	$-187 - (-111) - (+52)$	1
	$= -128 \text{ (kJ mol}^{-1}\text{)}$	1

19. M/J 22/P21/Q1/f(ii)

1(f)(ii)	M1 sketch shows exothermic reaction with a 'hump' AND labelled reactants ($\text{Mg} + \text{O}_2$) and products (MgO) M2 arrow from reactants / $\text{Mg} + \text{O}_2$ to products / MgO shown as ΔH M3 arrow showing activation energy / E_a / (+)148 	3
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20. M/J 22/P22/Q1/b(iii,iv)

1(b)(iii)	M1 (enthalpy change / energy change) when one mole of a compound / substance is formed M2 from its elements in their standard states	2
1(b)(iv)	M1 correct method to find amount Mg_3N_2 formed in reaction $3.645 \div 24.3 \text{ mol Mg produces } 0.15 \div 3 \text{ mol of } \text{Mg}_3\text{N}_2$ OR $0.05 \text{ mol of } \text{Mg}_3\text{N}_2$ M2 use M1 in correct expression to find energy released by 1 mol Mg_3N_2 $23.05 \div \text{M1}$ OR $23.05 \div \text{mol } \text{Mg}_3\text{N}_2$ OR $23\,050 \div \text{M1}$ OR $23\,050 \div \text{Mg}_3\text{N}_2$ M3 calculate value from M2 expression correctly $\Delta H = -(M2) = -461(.0) \text{ kJ mol}^{-1}$ OR $-461\,000 \text{ J mol}^{-1}$	3

21. M/J 22/P22/Q3/b

3(b)	evaluation, to give a value, based on calculation using: - all three correct bond energies [$\text{HI} = 299$, $\text{H}_2 = 436$, $\text{I}_2 = 151$] - correct use of stoichiometry to calculate ΔH for 1 mol HI $x = 1$ and $y = \frac{1}{2}$ - calculate a value for ΔH correctly i.e. $x(299) - y(436 + 151)$ $\Delta H = (+)5.5 \text{ kJ mol}^{-1}$	2
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22. M/J 22/P23/Q3/a(ii)

3(a)(ii)	M1 expression OR value derived from using all bond energy values and ΔH correctly, ignoring stoichiometry $-7 = (n350 + m360 + p410) - (q410 + r350 + s\text{C}\equiv\text{O})$. M2 correct calculation based on correct expression AND stoichiometry $[-7 = 2(350) + 2(360) + 10(410) - 10(410) - 350 - x] = (+)1077 \text{ kJ mol}^{-1}$	2
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23. MAR 22/P22/Q2/c(iii)

2(c)(iii)	$-1640 \text{ (kJ mol}^{-1}\text{)}$	1
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24. O/N 21/P21/Q3/a(ii)

3(a)(ii)		1
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25. O/N 21/P22/Q1/b

1(b)	(enthalpy / energy change) when one mole of a compound is formed from its elements in their standard states	2
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26. O/N 21/P23/Q3/a(ii)

3(a)(ii)		1
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27. MAR 21/P22/Q1/c(i)

1(c)(i)	M1: end higher than start AND 'hill' for E_a M2: E_a AND ΔH labelled	2
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28. MAR 21/P22/Q3/a(i)

3(a)(i)	M1: $\Delta H_f + (-2036) = 4 \times -384$ M2: $\Delta H_f = (+)500$	2
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29. O/N 20/P21/Q2/c(ii)

2(c)(ii)	M1: $\Delta H_f = -1434 - (-635 + -297)$ M2: $= -502 \text{ (kJ mol}^{-1}\text{)}$	2
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30. O/N 20/P23/Q2/c(ii)

2(c)(ii)	M1: $\Delta H_f = -1434 - (-635 + -297)$	2
	M2: $= -502 \text{ (kJ mol}^{-1}\text{)}$	

31. M/J 20/P22/Q3/d

3(d)(i)	M1 (enthalpy change) when 1 mole of sucrose M2 burns/combusts/reacts in excess air/oxygen OR completely burns/combusts/reacts in air/oxygen	2
3(d)(ii)	M1 for finding amount of energy released per gram of sucrose using $\Delta H/Q = mc\Delta T$ OR $\Delta H = -mc\Delta T$ $= (-)250 \times 4.18 \times (40.7 - 25) = (-)16406.5 \text{ J per gram OR } (-)16.4065 \text{ kJ/g}$ M2 for finding amount (mol) sucrose in 1g = $1/342 \text{ mol}$ M3 = $M1 / (M2 \times 1000)$ $\Delta H = -5610 \text{ kJ mol}^{-1}$ (3 sig figs) OR $-5611 \text{ kJ mol}^{-1}$ (4 sig figs)	3

32. MAR 20/P22/Q1/g(ii,iii,iv)

1(g)(ii)	M1 (enthalpy / energy change) when one mole of a compound / substance is formed M2 from its elements in their standard states	2
1(g)(iii)	$-(-602 + -188) + (\Delta H_f [\text{MgO}_2] + -286) = -96$ $\Delta H_f [\text{MgO}_2] = -600 \text{ (kJ mol}^{-1}\text{)}$	2
1(g)(iv)	$-(-600) - (+602) = -2 \text{ (kJ mol}^{-1}\text{)}$	1

33. O/N 19/P21/Q3/b(ii,iii,iv)

3(b)(ii)	M1 (enthalpy change when) 1 mol of a substance M2 EITHER burns / combusts / reacts in excess air / oxygen OR completely burns / combusts / reacts in air / oxygen	2
3(b)(iii)	M1 $m = 200$ and $\Delta T = 37.5 - 18.5$ M2 $Q = mc\Delta T = 200 \times 4.18 \times (37.5 - 18.5) = 15884 \text{ (J)}$	2
3(b)(iv)	M1 mol of thiophene used $= 0.63 / 84.1$ OR $7.49(1082045) \times 10^{-3}$ M2 calculation $\div 1000$ AND negative sign $\Delta H_c = \frac{-(iii)}{1000} \div n = \frac{-(iii)}{21000} \div (0.63 / 84.1)$ $= -2120 \text{ (-2120.39) (kJ mol}^{-1}\text{)}$	2

34. O/N 19/P23/Q3/b(ii,iii,iv)

3(b)(ii)	M1 (enthalpy change when) 1 mol of a substance M2 EITHER burns / combusts / reacts in excess air / oxygen OR completely burns / combusts / reacts in air / oxygen	2
3(b)(iii)	M1 $m = 200$ and $\Delta T = 37.5 - 18.5$ M2 $Q = mc\Delta T = 200 \times 4.18 \times (37.5 - 18.5) = 15884 \text{ (J)}$	2
3(b)(iv)	M1 mol of thiophene used $= 0.63 / 84.1$ OR $7.49(1082045) \times 10^{-3}$ M2 calculation $\div 1000$ AND negative sign $\Delta H_c = \frac{-(iii)}{1000} \div n = \frac{-(iii)}{21000} \div (0.63 / 84.1)$ $= -2120 \text{ (-2120.39) (kJ mol}^{-1}\text{)}$	2

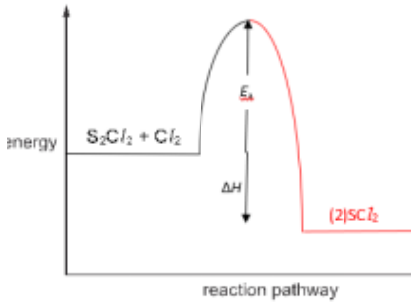
35. M/J 19/P21/Q2/e

2(e)	M1 Si—H bond is (much) weaker than C—H bond	1
	M2 low activation energy <i>ORA</i>	1

36. MAR 19/P22/Q2/c(iii)

2(c)(iii)	M1 labelled reactants AND products lower on right	2
	M2 labelled enthalpy change with correct arrow	

37. O/N 18/P22/Q2/b(ii)

2(b)(ii)	 M1 profile for exothermic reaction [1] M2 identification of ΔH and E_a [1]	2
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38. O/N 18/P22/Q3/a

3(a)	$\Delta H_f = (-692.9) + 3(-61.8) - (-182.1) - 3(-204.6)$ $= -82.4 \text{ (kJ mol}^{-1}\text{)}$ M1 $\Delta H_f = x(-692.9) + y(-61.8) - v(-182.1) - w(-204.6)$ where x, y, v and w are integers ≥ 1 [1] M2 use of correct stoichiometry where $x=1, y=3, v=1$ and $w=3$ [1] M3 -82.4 [1]	3
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39. M/J 18/P21/Q1/b

1(b)	$-196 + 6\text{S}=\text{O} = (4 \times 534) + 496$	1
	$\text{S}=\text{O} = 2828 / 6 = 471(.3)$	1

40. O/N 17/P21/Q1/a

1(a)(i)	energy needed / required to break a mole of (covalent) bonds	1
	(All) in the gaseous state	1
1(a)(ii)	$-92 = \{944 + 3(436)\} - 6E(\text{N-H})$	1
	$E(\text{N-H}) = (+)390.7 / 390.67 / 391$	1

41. O/N 17/P22/Q2/a

2(a)	-444	1
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42. O/N 17/P23/Q1/a

1(a)(i)	energy needed / required to break a mole of (covalent) bonds	1
	(All) in the gaseous state	1
1(a)(ii)	$-92 = \{944 + 3(436)\} - 6E(\text{N-H})$	1
	$E(\text{N-H}) = (+)390.7 / 390.67 / 391$	1

43. M/J 17/P22/Q3/a

3(a)	(+) 103	1
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44. M/J 17/P23/Q3/a

3(a)(i)	(enthalpy / energy change) when one mole of a compound is formed	1
	from its elements in their standard states / standard conditions	1
3(a)(ii)	$(\Delta H_f = \sum \Delta H_f \text{ products} - \sum \Delta H_f \text{ reactants})$ $-196 = 2\Delta H_f \text{ SO}_3 - (2 \times -296.8)$ $2\Delta H_f \text{ SO}_3 = -196 + (2 \times -296.8) = -789.6$	1
	$\Delta H_f \text{ SO}_3 = -394.8 \text{ (kJ mol}^{-1}\text{)}$	1

45. MAR 17/P22/Q2/c(i)

2(c)(i)	$\Delta_r H = \Delta_r H\{\text{products}\} - \Delta_r H\{\text{reactants}\} = 2 \times (-242) - 4 \times (-92)$	1
	$= -116$ (sign AND answer)	1

46. O/N 16/P21/Q2/a,b,c

2(a)(i)	Enthalpy/energy/heat change when one mole of a substance Burns/combusts/reacts in excess oxygen OR Completely burns/combusts/reacts in oxygen under standard conditions	1 1 1	3
2(a)(ii)	$\text{C}_2\text{H}_5\text{OH} + 3\text{O}_2 \rightarrow 2\text{CO}_2 + 3\text{H}_2\text{O}$	1	1
2(b)(i)	6813.4/6813/6810/6800 (J)	1	1
2(b)(ii)	-1362.68/-1362.7/-1363/-1360/-1400 (kJ)	1	1
2(b)(iii)	Any 2 from: heat/energy losses (to air and/or to the container/surroundings) incomplete combustion (volatile) ethanol evaporated ethanol is impure not all energy is lost as heat	1 1	2
2(c)(i)	$ \begin{array}{c} 3\text{C(s)} + 4\text{H}_2\text{(g)} + \frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{C}_3\text{H}_7\text{OH(l)} \\ \swarrow \quad \searrow \quad \swarrow \\ 3(-393.5) \quad 4 \times (-285.8) \quad -2021.0 \\ \searrow \quad \swarrow \quad \swarrow \\ 3\text{CO}_2 + 4\text{H}_2\text{O} \end{array} $	1+1 1	3
2(c)(ii)	$\Delta H_f + (-2021.0) = 3(-393.5) + 4(-285.8)$ $\Delta H_f = -302.7 \text{ (kJ mol}^{-1}\text{)}$	1 1	2

51. O/N 15/P21/Q2/a,b(i)

2 (a) (i)	The enthalpy change when one mole of a compound is formed from its element(s)	[1] [1]	[2]
(ii)	$\text{S(s)} + 1\frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{SO}_3\text{(l)}$	[1]	[1]
(b) (i)	$944 + (3 \times 436) = 2252$ $6 \times 390 = 2340$ $2252 - 2340 = -88 \text{ (kJ mol}^{-1}\text{)}$	[1] [1] [1]	[3]

52. O/N 15/P22/Q2/a,b

2 (a)	M1 Heat (energy) change (or $H_{\text{prod}} - H_{\text{react}}$) measured at constant pressure OR enthalpy change when the amount/moles of reactants as shown in a (reaction) <u>equation</u> react together to give products M2 measured at standard conditions	[1] [1]	[2]
(b) (i)	$q = 2125.53$	[1]	[1]
(ii)	amount = 0.025(0)	[1]	[1]
(iii)	-85.(0)	[1]	[1]
(iv)	$ \begin{array}{ccc} & (\text{MgSO}_4\text{(s)} + 7\text{H}_2\text{O(l)} \rightarrow \text{MgSO}_4 \cdot 7\text{H}_2\text{O(s)}) & \\ & \swarrow \quad \searrow & \\ -85.0 \text{ (kJ mol}^{-1}\text{)} & & (+)9.60 \text{ (kJ mol}^{-1}\text{)} \\ & \searrow \quad \swarrow & \\ & \text{MgSO}_4\text{(aq)} & \end{array} $	[1]	[1]
(v)	$\Delta H + 9.6 = -85.0$ $\Delta H = -85.0 - 9.6 = -94.6 \text{ (kJ mol}^{-1}\text{)}$	[1]	[1]

53. O/N 15/P23/Q2/a,b(i)

2 (a) (i)	The enthalpy change when one mole of a compound is formed from its element(s)	[1] [1]	[2]
(ii)	$\text{S(s)} + 1\frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{SO}_3\text{(l)}$	[1]	[1]
(b) (i)	$944 + (3 \times 436) = 2252$ $6 \times 390 = 2340$ $2252 - 2340 = -88 \text{ (kJ mol}^{-1}\text{)}$	[1] [1] [1]	[3]

54. M/J 15/P21/Q3/a

3 (a)	Bond breaking = C=O = 740 C-H = 410 = 1150 kJ Bond forming = C-C = 350 C-O = 360 O-H = 460 = 1170 kJ Enthalpy change = 1150 - 1170 = -20 kJ mol ⁻¹	[1] [1] [1]	[3]
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55. M/J 15/P22/Q3/a(i)

3 (a) (i)	Bond breaking = $\text{Cl-Cl} = 242$ $\text{C-H} = 410 = 652 \text{ kJ}$ Bond forming = $\text{C-Cl} = 340$ $\text{H-Cl} = 431 = 771 \text{ kJ}$ Enthalpy change = $652 - 771 = -119$	[1] [1] [1]	[3]
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